

User Guide

Football Performance Analysis Platform | GPS, physical load and tactical intelligence

This tool turns athletes' **GPS and inertial sensor** data (collected via Catapult Connect) into **clear, actionable insights** on physical performance and tactical behaviour. In just a few clicks you go from raw spreadsheets to **heat maps, load profiles, worst-case scenarios (WCS) and collective tactical visualisations** — including a 3D replay of the match. The goal: to give coaching and performance staff a **fast, visual and scientifically grounded** reading of each athlete and of the team as a system.

Getting started in 4 steps

1

Connect

In the sidebar, enter your Connect **Token** and load the athletes and activities.

2

Choose the activity

Select the **activity** (training or match) you want to analyse from the loaded list.

3

Select period(s) and athletes

Pick one or more **periods** (1st half, 2nd half, etc.) and the **athletes** of interest. With several periods, the tool automatically creates a combined view.

4

Explore the 5 tabs

Data is organised into five modules. Each tab has interactive charts (hover for details, drag to zoom) and export buttons.

The 5 platform modules

SUMMARY

- **Overview:** total distance, PlayerLoad, top speed and key session indicators.
- **By Position:** compares athletes by tactical role (defender, midfielder, forward...).

FIELD & GPS

- **Football Field:** occupancy heat maps and trajectory — this is where you mark the field.
- **Match Story:** heat maps by phase, showing how presence evolves over time.
- **WCS:** the most demanding passages of the match by rolling window.

PHYSICAL LOAD

- **Efforts:** accelerations, decelerations and sprints, with zone and direction.
- **Time Windows:** rolling-window intensity and high / medium-high peaks.
- **Neuromuscular, Acc-Vel and HR:** fatigue, acc×speed profile and cardiac load (TRIMP + zones).

COLLECTIVE TACTICS

- The team as a **system:** cross-references every athlete's position at the same instant.
- Four views: **Pitch Control, Team Breathing, Voronoi** and **3D Replay**.
- Works even with **GPS-only** devices (reconstructs the coordinates).

Marking the field: enabling location data

Raw GPS is just a cloud of **latitude/longitude** points. For the app to understand *where* on the pitch each point is — and thus produce zone occupancy, field thirds, directional efforts and the whole **Collective Tactics** module — you must tell it **where the field is and how it is oriented**. This is called **marking (or calibrating) the field**, and it is done once per venue, over a satellite image.

Where it is

Tab **Field & GPS > Football Field**. Select a **reference athlete who has GPS** in the session (the field belongs to the venue, not the athlete — anyone with signal works). If the session has GPS points, a **satellite map** appears with the point cloud and a **yellow target** marking the field centre.

Step-by-step marking

1

Show the field

Click **Show Field** to overlay the field drawing (white lines) on the satellite image.

2

Centre it

Drag the **yellow target** — or edit **Lat/Lon** — until the drawing covers the GPS point cloud, over the real pitch in the image.

3

Rotate

Use the **Rot** buttons (« =-10°, < =-1°, > =+1°, » =+10°) until the **drawing's lines match the real lines** of the field on the satellite. This is the most important step for tactical fidelity.

4

Adjust dimensions

Set **Length** (90–120 m) and **Width** (55–90 m) to match the real size. Default is 105×68 m. **Marg.** adds an outer margin beyond the lines if you wish.

5

Apply

Click **Apply Field**. From then on, location coordinates feed the heat maps, directional efforts and the whole **Collective Tactics** module.

Control in the map panel	What it does
Show Field	Toggles the field drawing over the satellite.
Lat / Lon	Moves the field centre. Drag the yellow target or edit the numbers.
Rot « < > »	Rotates the field: « -10°, < -1°, > +1°, » +10°. Align the drawing's lines with the pitch.
Length / Width	Length (90–120 m) and width (55–90 m). Default 105×68 m.
Marg.	Extra margin drawn outside the lines (0–15 m).
Apply Field	Confirms the setup and loads the field coordinates.

Save for everyone (venue database)

Right after applying, the app offers to **save the configuration to the shared venue database**, with a name (e.g. “Municipal Stadium”). It's well worth it: next time **any user** opens an activity **at this same venue**, the field is already **configured automatically** — no need to mark it again. You'll see the notice “Field loaded from the shared database”.

- **Readjust**: redo the marking from scratch (back to the satellite map).
- **Venues**: opens the manager to list, update or delete saved venues.

How to tell it's well marked

- The **white drawing lines** match the field lines in the satellite image.
- The **GPS point cloud** stays inside the field, without spilling far outside.
- The **goals** sit at the correct ends and the halfway line crosses the real centre circle.
- Tip: start with **rotation**, then adjust the **centre**, and finally the **dimensions**.

Spotlight: the Collective Tactics tab

It's the most innovative module. By syncing the position of all selected athletes, it reveals the team's behaviour as a whole — impossible to see athlete by athlete. It requires **at least 2 athletes with position** in the same period. The four views:

Pitch Control

- William Spearman's model (Liverpool FC). Colours the field by **space dominance** — who would reach each point first, accounting for speed and direction.

Team Breathing

- Shows the block **compressing** when defending and **expanding** in possession, with width, length, area and spread over time.

Voronoi

- The tactical “stained glass”: each region of the field belongs to the nearest player, exposing **space coverage and gaps** in the structure.

3D Replay

- A navigable **synthetic broadcast** (rotate the camera, zoom) with striped grass, goals and markings — reconstructed from coordinates alone.

Animation controls

- **Analysis window**: from 30 s to 10 min, a custom duration or the **whole period**. Short windows show continuous movement (real time); the whole period gives the macro view of the entire match (with more spaced-out frames).
- **Window start**: drag it to place the analysis at **any moment** of the match (including the final minutes).
- **Speed**: **1×** = **real time** of the match; below is slow motion, above is sped up. Use **Play/Pause** and the slider to navigate the frames.

Two time controls — don't confuse them

- The **Window start** slider (above the chart) **moves the window** across the whole match.
- The slider **inside the chart** only scrubs the **frames of the current window**.

Troubleshooting (common difficulties)

P: The map doesn't appear / "No real GPS point found".

R: That athlete's sensor didn't get a GPS *lock* in the session. Since the field belongs to the venue, pick **another reference athlete** with GPS signal to do the marking.

P: The field is crooked or misaligned on the satellite.

R: First adjust the **rotation** (« < > ») and then the **centre** (Lat/Lon) until the drawing's lines match the pitch. Also check **Length/Width** — wrong dimensions distort everything.

P: Collective Tactics shows "coordinates reconstructed from GPS / approximate alignment".

R: Your device is **GPS-only** (no native x/y). The **relative positions** between players are faithful; for better absolute alignment, **mark the field** in Field & GPS — it will be used automatically here.

P: "This tab needs at least 2 athletes with position in the same period".

R: In the sidebar, select **2 or more athletes** that have GPS in that period.

P: The animation looks like "photos" jumping, with no continuous motion.

R: Long passages have spaced-out frames. Reduce the **Analysis window** to **1–2 min** and use the **Window start** to focus on the play. The whole period is for the macro view.

P: GPS distances differ from what I expected.

R: Check that the field is well marked and with the **real dimensions** of the pitch. The app also compares the field-filtered distance with the integrated GPS — big differences point to a marking that needs review.

P: I configured the field but want to redo it.

R: Use **Readjust** (back to the map) or **Venues** to update/delete the saved venue.

Tips to get the most out of it

- **Save the venue** after marking the field: it benefits the whole team automatically.
- **Combine periods** to see the whole match on a continuous timeline, without half-time.
- **Export** tables and events to CSV for reports and longitudinal tracking.
- To study a play in real time, use a **1–2 min** window and aim the start at the exact moment.
- Hover over the charts to see **exact values**; drag to zoom.

Additions, suggestions, comments

Have ideas to improve the tool, found something to adjust, or want to suggest a new feature? Your feedback is very welcome. Write to:

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